Genuine Multipartite Entanglement Analysis — Molecular System

MOLECULE

NH2- (NH2-)
Basis set   : cc-pVDZ
Charge   : -1
Multiplicity : 1
Total e-   : 10
n-body order : 3

GROUND STATE ENERGIES

E0 (Ha) : -55.48749269
E1 (Ha) : -55.35926382
Gap (Ha) : 0.12822887
HF (Ha) : -55.48003660

CONVEX-ROOF METHODS

Methods run   : gcg
Restarts   : 8

C_LB : coherence witness bound
C_expo: unitary opt (exp map)
C_stf : Stiefel manifold (Alg. B)
C_UB : eigendecomposition bound

ACTIVE SPACE CAS(6,5)

Frozen core orbs : 2
Frozen e-   : 4
Active e-   : 6
Active orbs   : 5
Spin-orbitals   : 10
Hilbert dim   : 100

SPIN PROPERTIES

<S2>   : 0.000000
S   : 0.0000
<Sz>   : 0.000000
Multiplicity : 1
Spin state   : Singlet (S=0)

GME CRITERIA

C_GME = sqrt(2 * min_A (1 - Tr(rho_A^2)))

Convex roof (mixed):
  C_GME(rho) = min Sum p_i C_GME(|psi_i>)

Bounds ordering:
  C_LB <= C_GME <= C_UB

DOUBLE OCCUPANCY

<D> total : 2.992471
<D> / site : 0.598494

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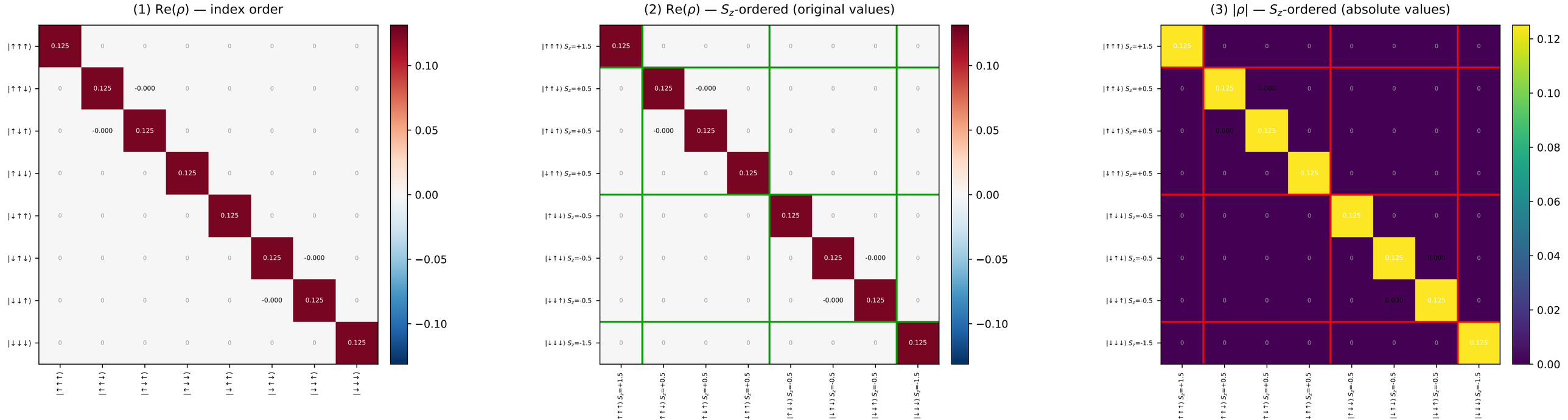
Body order : 3
Clusters   : 10

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Summary — 3-Body RDMs (NH2-, CAS(6,5))

Orbitals	Purity	Entropy	C_GME (pure)	C_LB	C_gcg	C_UB	State
0,1,2	0.125000	2.079441	—	0.000000	0.000265	0.000000	STRONGLY MIXED
0,1,3	0.302464	1.492167	—	0.000000	0.000500	0.456706	MIXED
0,1,4	0.428599	1.040659	—	0.000000	0.227730	0.573683	MIXED
0,2,3	0.130564	2.054887	—	0.000000	0.000116	0.000609	STRONGLY MIXED
0,2,4	0.235332	1.713302	—	0.000000	0.000000	0.001303	MIXED
0,3,4	0.495734	0.721811	—	0.000000	0.587580	0.621594	MIXED
1,2,3	0.255909	1.646852	—	0.000000	0.001699	0.001403	MIXED
1,2,4	0.126994	2.071023	—	0.000000	0.000451	0.000000	STRONGLY MIXED
1,3,4	0.492355	0.744579	—	0.000000	0.547454	0.619952	MIXED
2,3,4	0.226664	1.741217	—	0.000000	0.000906	0.000000	MIXED

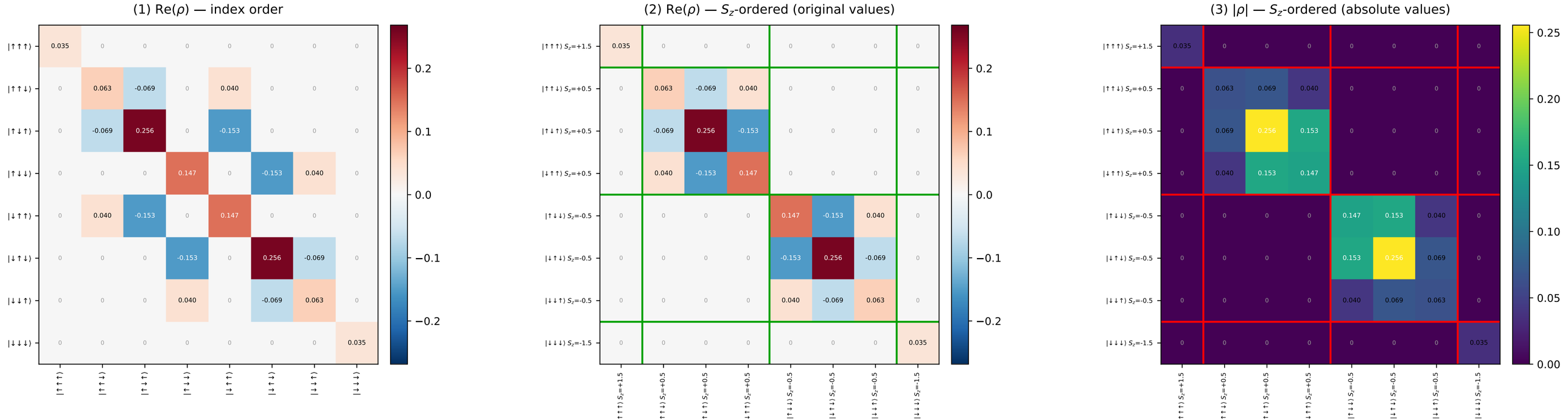
3-Body RDM (Sites 0,1,2) — same matrix in all three views



Trace=1.000000 Purity=0.125000 Entropy=2.079441 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000265 ≤ C_{UB} = 0.000000 verdict: GME

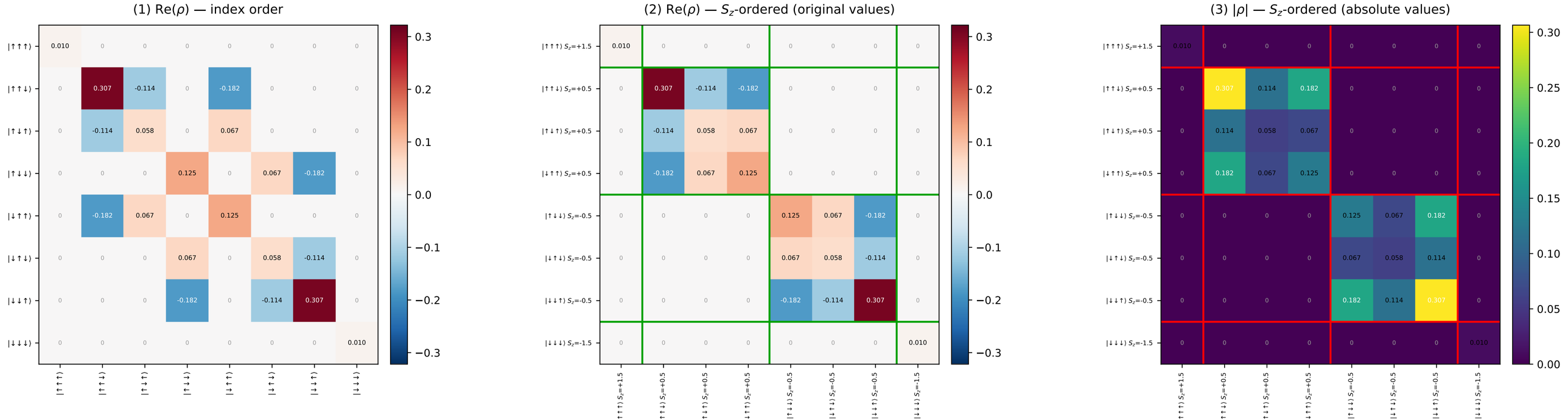
3-Body RDM (Sites 0,1,3) — same matrix in all three views



Trace=1.000000 Purity=0.302464 Entropy=1.492167 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000500 ≤ C_{UB} = 0.456706 verdict: GME

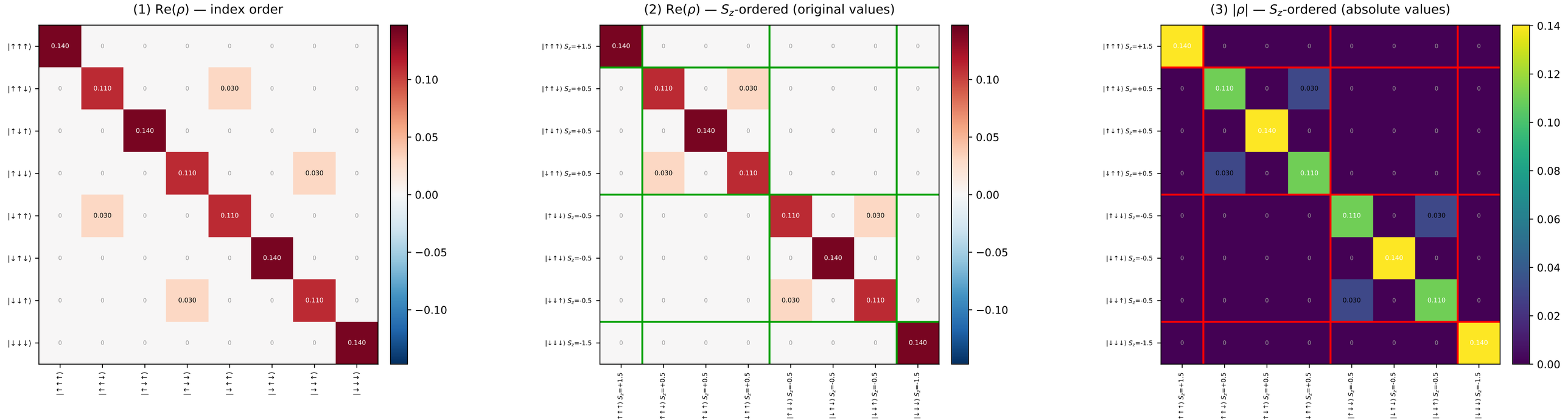
3-Body RDM (Sites 0,1,4) — same matrix in all three views



Trace=1.000000 Purity=0.428599 Entropy=1.040659 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.227730 ≤ C_{UB} = 0.573683 verdict: GME

3-Body RDM (Sites 0,2,3) — same matrix in all three views

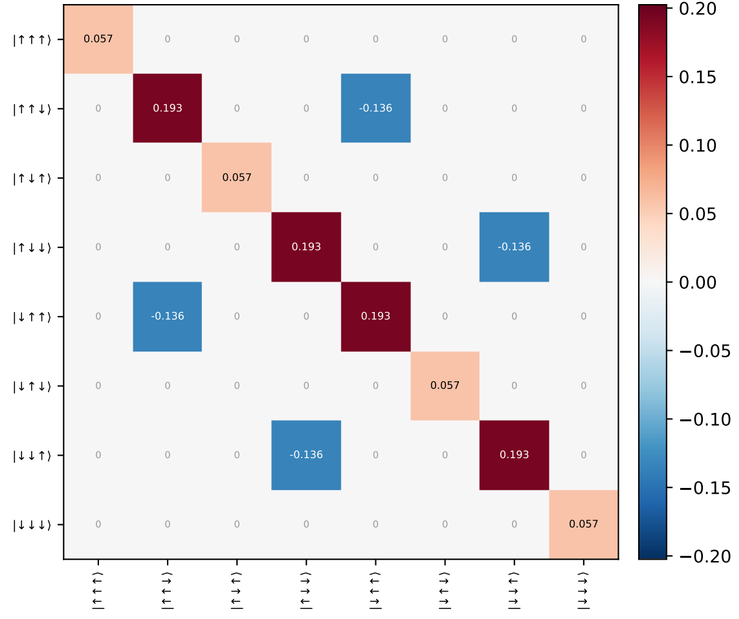


Trace=1.000000 Purity=0.130564 Entropy=2.054887 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

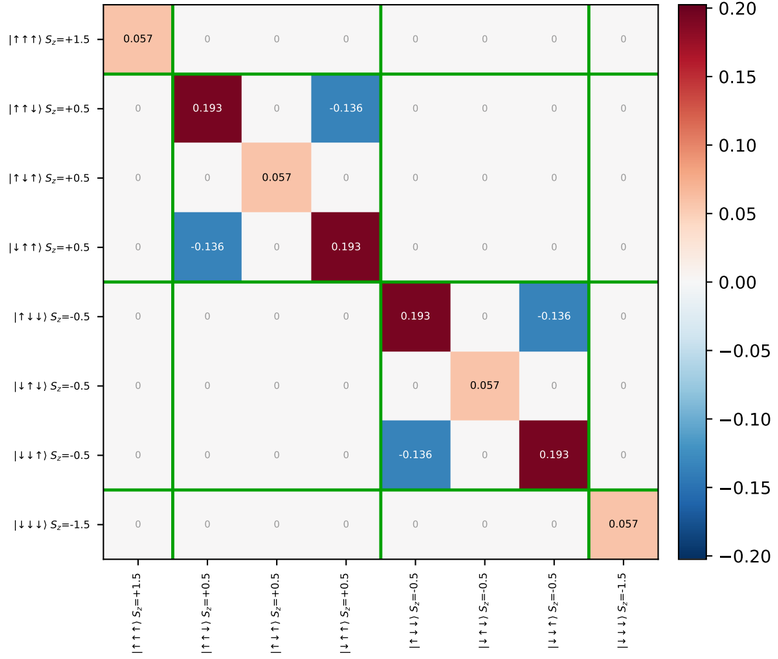
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000116 ≤ C_{UB} = 0.000609 verdict: GME

3-Body RDM (Sites 0,2,4) — same matrix in all three views

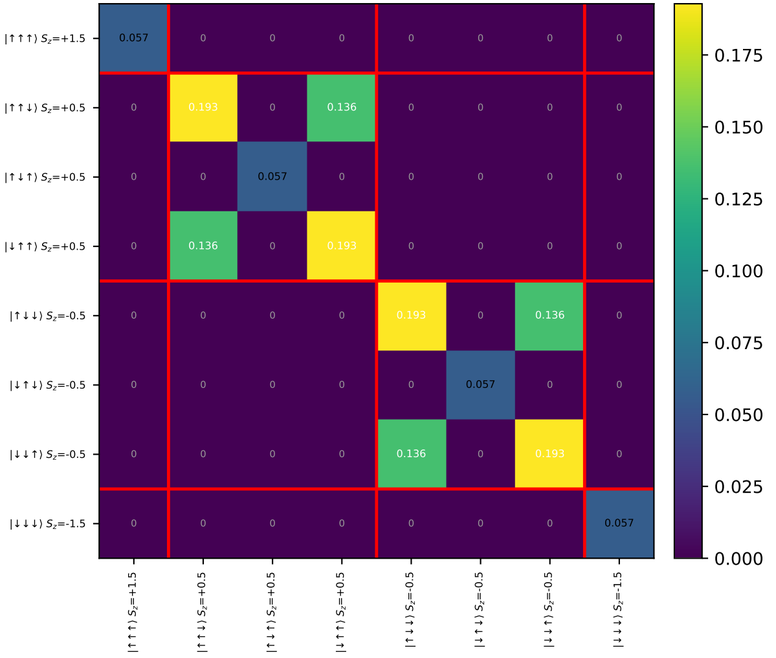
(1) Re(ρ) — index order



(2) Re(ρ) — S_z-ordered (original values)



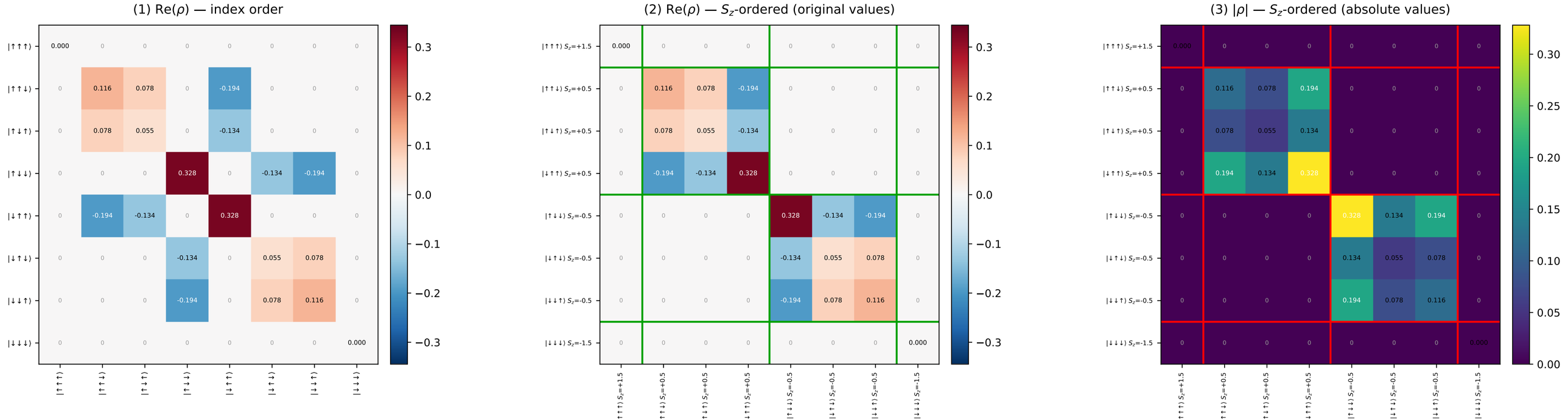
(3) $|\rho|$ — S_z-ordered (absolute values)



Trace=1.000000 Purity=0.235332 Entropy=1.713302 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

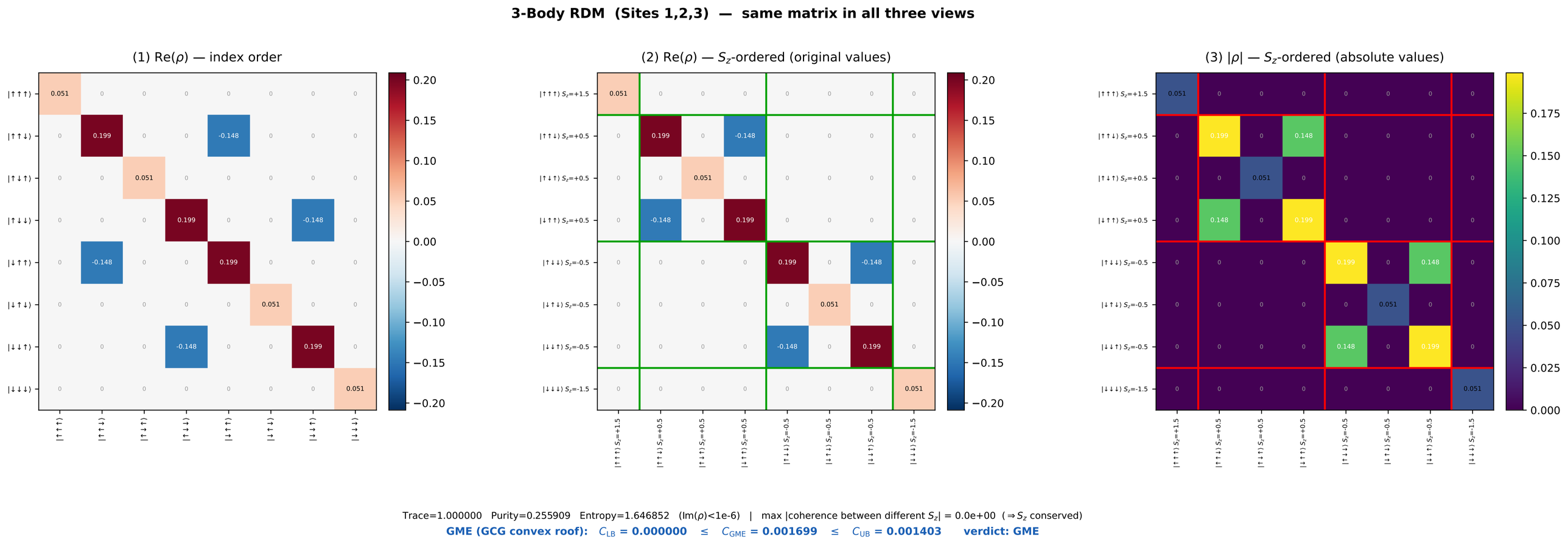
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000000 ≤ C_{UB} = 0.001303 verdict: no GME

3-Body RDM (Sites 0,3,4) — same matrix in all three views

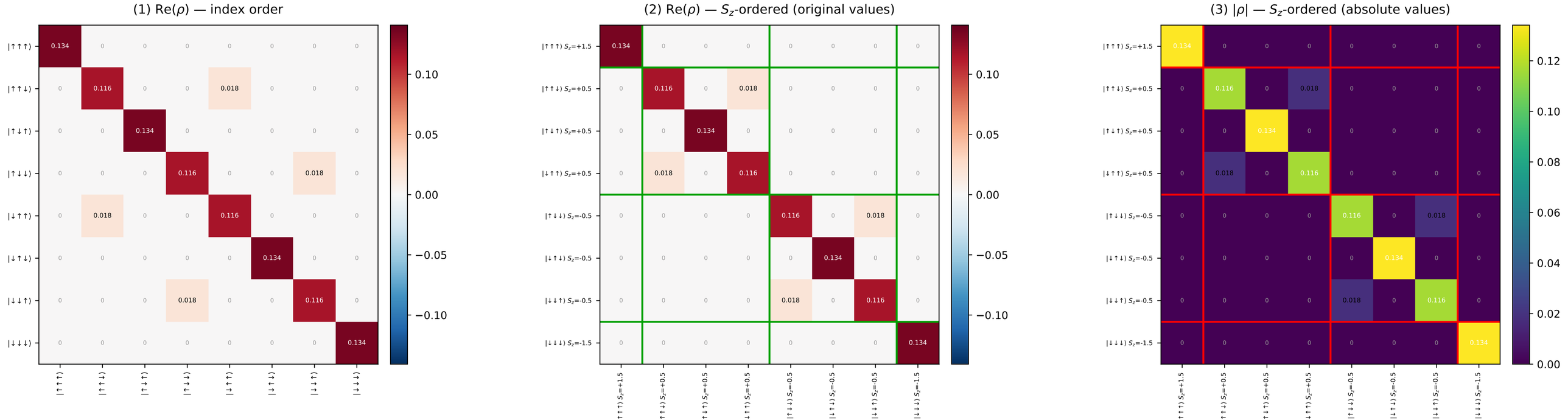


Trace=1.000000 Purity=0.495734 Entropy=0.721811 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.587580 ≤ C_{UB} = 0.621594 verdict: GME


3-Body RDM (Sites 1,2,3) — same matrix in all three views
(1) Re(ρ) — index order
(2) Re(ρ) — S_z-ordered (original values)
(3) |ρ| — S_z-ordered (absolute values)
Trace=1.000000 Purity=0.255909 Entropy=1.646852 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.001699 ≤ C_{UB} = 0.001403 verdict: GME

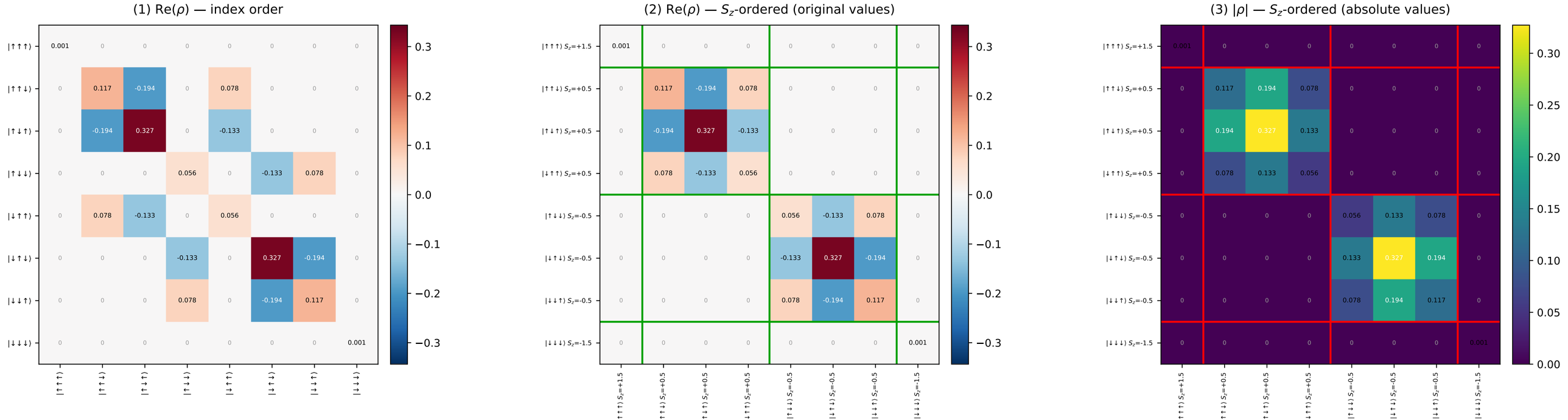
3-Body RDM (Sites 1,2,4) — same matrix in all three views



Trace=1.000000 Purity=0.126994 Entropy=2.071023 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000451 ≤ C_{UB} = 0.000000 verdict: GME

3-Body RDM (Sites 1,3,4) — same matrix in all three views

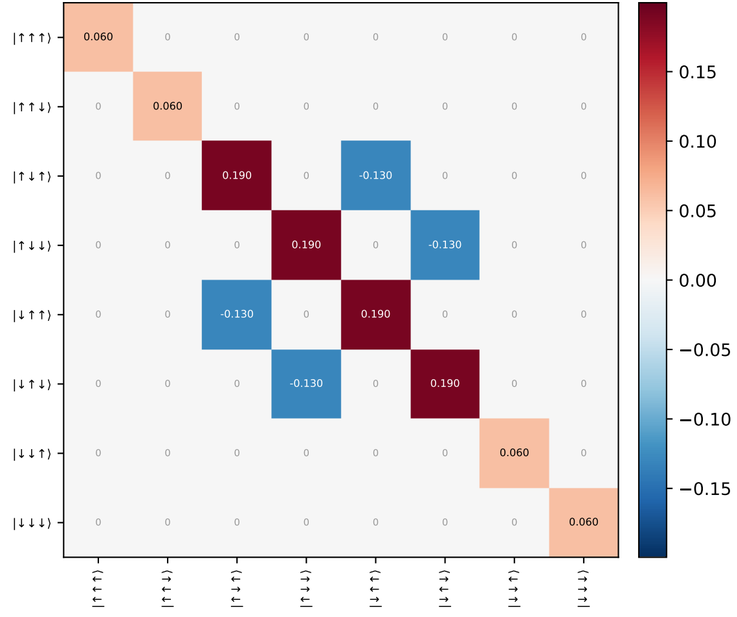


Trace=1.000000 Purity=0.492355 Entropy=0.744579 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 (⇒ S_z conserved)

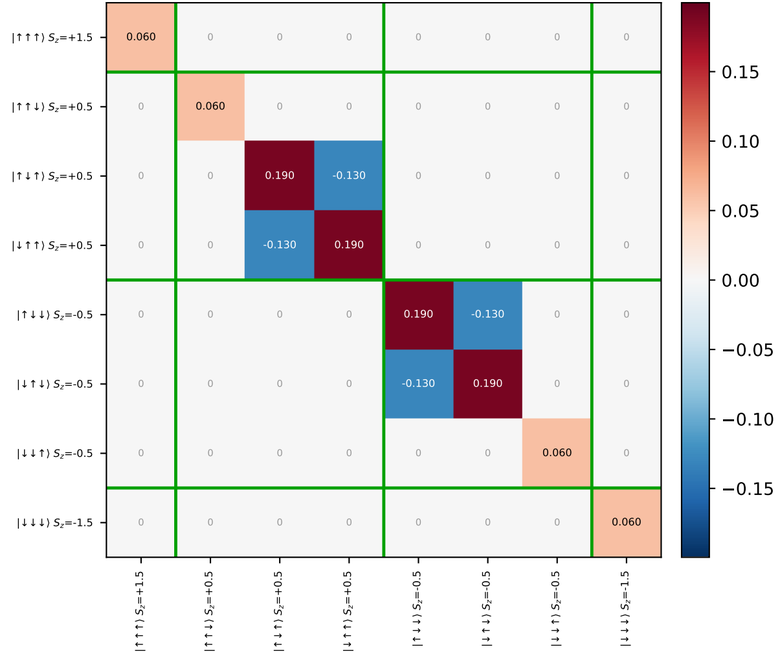
GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.547454 ≤ C_{UB} = 0.619952 verdict: GME

3-Body RDM (Sites 2,3,4) — same matrix in all three views

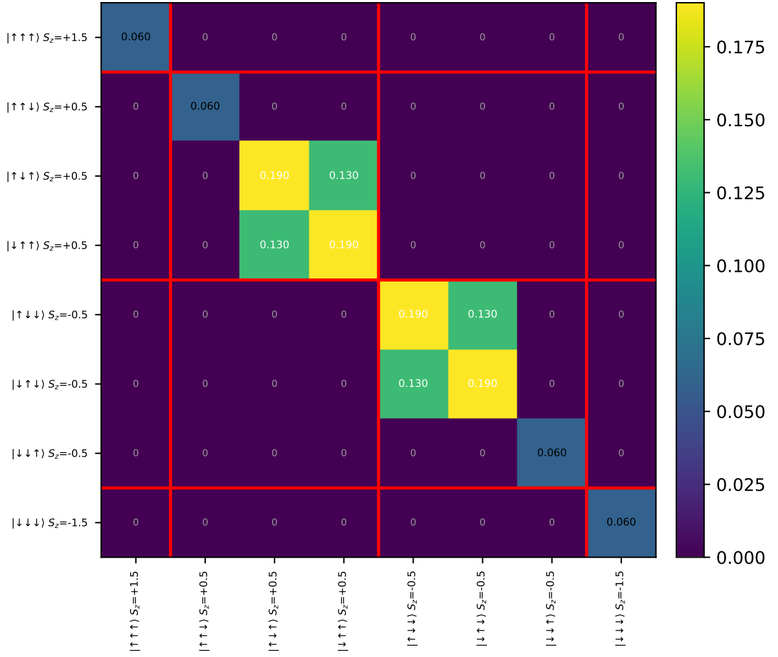
(1) Re(ρ) — index order



(2) Re(ρ) — S_z-ordered (original values)



(3) $|\rho|$ — S_z-ordered (absolute values)



Trace=1.000000 Purity=0.226664 Entropy=1.741217 (Im(ρ)<1e-6) | max |coherence between different S_z| = 0.0e+00 ($\Rightarrow S_z$ conserved)

GME (GCG convex roof): C_{LB} = 0.000000 ≤ C_{GME} = 0.000906 ≤ C_{UB} = 0.000000 verdict: GME